SEMINAR REPORT

A seminar Report on

AI Generated Text Detection

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960248	388554	Mansi Tulshidas Patil
2	24610960249	388555	Prachi Nilesh Patil
3	24610960276	388577	Jagruti Baliram Shinde
4	25610960010	388608	Nikita Laxman Patil

FOR THE ACADEMIC YEAR: 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code: 0059)

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960248	388554	Mansi Tulshidas Patil
2	24610960249	388555	Prachi Nilesh Patil
3	24610960276	388577	Jagruti Baliram Shinde
4	25610960010	388608	Nikita Laxman Patil

of **CO5K** Semester of Diploma in **Computer Engineering** of **S.S.V.P.S's B. S. Deore Polytechnic, Dhule** (Institute Code: 0059) has completed the term work satisfactorily in Course **Seminar Presentation and Project Initiation (Course Code: 315003)** for the academic year **2026 to 2027** as prescribed in the curriculum.

Place : Dhule

Date : 17/10/2026

Signature:

Name : Mr. Nishad D. Patel

Subject Teacher

Signature:

Name : Mr. Nishad D. Patel

Head of the Department

Seal of Institution

TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2 - 4
3	Information/Chapters	5 - 8
4	Advantages and Disadvantages	9
5	Conclusion	10
6	Project Initiation:	11
	6.1) Description of problem statement.	11
	6.2) Scope and objectives.	11
	6.3) Stake holder	11
	6.4) Platform/ Equipment/ Resources identification.	12
7	Seminar Presentation	13 - 22
	Bibliography	23
	References	24

ABSTRACT

The rapid growth of Large Language Models (LLMs) such as ChatGPT has made it very easy to generate fluent, human-like text. Although these tools are useful, they also create serious concerns in education, journalism, research and online content, such as plagiarism, misinformation and loss of authenticity. AI Generated Text Detection is the task of automatically deciding whether a given piece of text was written by a human or produced by a machine.

This seminar report presents a review of twelve research papers published between 2023 and 2026 on AI generated text detection. The reviewed work covers classical machine learning classifiers such as Logistic Regression, Support Vector Machine, Naive Bayes and XGBoost, transformer-based models such as BERT, ensemble and domain-gating approaches, explainable AI techniques for academic integrity, and survey articles describing the advances and challenges of the field.

The report explains the working principle, system architecture, techniques and technologies, advantages and limitations of AI text detection systems. It concludes with a project initiation plan for developing a detector that can help in identifying AI generated text and supporting academic integrity.

LIST OF FIGURES

Fig. No.	Title	Page No.
3.4	Working of AI Generated Text Detection	6
3.5	Architecture of AI Generated Text Detection System	7

LIST OF TABLES

Table No.	Title	Page No.
2.2	Literature Review	4
6.1	Resources	12

CHAPTER 1
INTRODUCTION

Artificial Intelligence (AI) is a rapidly growing field of computer science that focuses on developing machines and systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, problem-solving, and language understanding. Machine Learning (ML), a major branch of AI, enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task. Deep Learning (DL), a subset of machine learning, uses multi-layered neural networks to learn complex patterns from large amounts of data.

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, analyze, and generate human language. NLP plays an important role in applications such as machine translation, sentiment analysis, text summarization, chatbots, and text classification. Recent advancements in NLP, particularly the development of Transformer-based architectures and Large Language Models (LLMs), have significantly improved the ability of machines to generate human-like text.

Large Language Models (LLMs), such as ChatGPT, Gemini, and Claude, can now generate essays, answers, news articles, code comments, and emails that closely resemble human writing. These tools are widely used by students, teachers, researchers, writers, and businesses because they save time, improve productivity, and assist in various writing and communication tasks.

AI-Generated Text Detection is an emerging research area that combines NLP, machine learning, and deep learning techniques to identify whether a piece of text is human-written or AI-generated. These features may include word distribution, sentence structure, writing style, text perplexity, and patterns in language usage.

Although AI-generated text detection has made significant progress, reliable detection remains challenging because language models continue to improve, generated text can be edited or paraphrased, and human-written and AI-generated texts may share similar linguistic characteristics. Therefore, developing accurate, robust, and interpretable detection methods is an important research challenge in the fields of Artificial Intelligence, Machine Learning, and Natural Language Processing.

CHAPTER 2
LITERATURE REVIEW

- **Review of Research Paper**

1. Enhancing AI Text Detection with Frozen Pretrained Encoders and Ensemble Learning

(Pudasaini, Miralles-Pechuán, Lillis and Llorens Salvador, 2025)

This work proposes an approach for detecting AI generated text using frozen pretrained encoders, which act as fixed feature extractors without being retrained, together with ensemble learning. Combining several models makes the detection more robust while keeping the training cost low.

2. Domain Gating Ensemble Networks for AI-Generated Text Detection

(Tripathi, Jin, Zhang, Dugan, Gao, Zhang, Zhao, Huan and Callison-Burch, 2025)

This work introduces a Domain Gating Ensemble Network in which several detectors are combined and a gating mechanism decides the importance of each detector according to the domain of the input text. The aim is to keep detection accurate when the text comes from different domains.

3. Research on Performance Optimization of BERT Model in AI-Generated Text Detection

(Jiarui Yu, 2026)

This work studies how the BERT model can be optimized for AI generated text detection. It focuses on improving the training and configuration of the transformer so that human written and machine generated text are classified more accurately.

4. AI Generated Text Detection

(Alikhanov, Akhmetzhan, Amangeldi, Polat, Demeubay, Zharas and Moldakhmetov, 2026)

This paper studies the detection of AI generated text using NLP and machine learning techniques and examines how well the models can separate human writing from machine writing.

5. AI-Generated Text Detection: A Review

(Lalit, Bhutani, Verma and Jain, 2023)

This review paper surveys the existing techniques for detecting AI generated text. It summarizes the main methods, discusses their limitations and highlights open challenges for future research.

6. Human vs Machine: An Automated Machine-Generated Text Detection Approach

(Jwaid, Roy, Pal, Debnath, Das and Bandyopadhyay, 2024)

This work presents an automated approach for detecting machine generated text. The system learns to tell apart text written by humans from text produced by language models.

7. Detecting AI-Generated Text in Educational Content: Leveraging Machine Learning and Explainable AI for Academic Integrity

(Najjar, Ashqar, Darwish and Hammad, 2023)

This work applies machine learning together with Explainable AI (XAI) to detect AI generated text in educational content. The use of explainability helps teachers understand why a text is flagged, which supports academic integrity.

8. AI-Generated Text Detection Based on Machine Learning Models

(Moran Zeng, 2024)

This work uses traditional machine learning classifiers such as Logistic Regression, Support Vector Machine and Naive Bayes to classify text as human written or AI generated.

9. Detecting AI-Generated Text Through NLP and Machine Learning Methodologies

(Zenith Malveen, 2024)

This work explores NLP and machine learning methodologies for detecting AI generated text, covering text pre-processing, feature extraction and classification.

10. Detecting AI Generated Text Based on NLP and Machine Learning Approaches

(Nuzhat Noor Islam Prova, 2024)

This work detects AI generated text with NLP and machine learning approaches and compares models such as XGBoost, Support Vector Machine and BERT.

11. Distinguishing Human From Machine: A Review of Advances and Challenges in AI-Generated Text Detection

(Fariello, Fenza, Forte, Gallo and Marotta, 2025)

This review discusses recent advances in AI generated text detection and the main challenges, such as new and improving language models, different domains and the reliability of detectors.

12. Research on AI-Generated Text Detection Based on Machine Learning Models

(Moran Zeng, 2024)

This research investigates the use of machine learning models for separating AI generated text from human written text by training classifiers on labelled text data and comparing their results.

Table No 2.1: Literature review:

Sr. No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
1	Pudasaini et al., 2025	Ensemble of frozen encoders	Frozen pretrained encoders; ensemble learning	Detect AI text with good accuracy at low training cost
2	Tripathi et al., 2025	Domain-aware ensemble	Domain Gating Ensemble Network (gating + multiple detectors)	Keep detection accurate across different text domains
3	Yu, 2026	Transformer optimization	BERT; model tuning and optimization	Improve BERT performance for AI text detection
4	Alikhanov et al., 2026	NLP and ML based detection	NLP techniques; machine learning models	Distinguish human and AI generated text
5	Lalit et al., 2023	Review	Survey of existing detection methods	Summarize techniques and challenges
6	Jwaid et al., 2024	Automated detection	Machine learning classification of machine-generated text	Separate human and machine text automatically
7	Najjar et al., 2023	ML with Explainable AI	Machine learning classifiers; explainable AI	Detect AI text in education and support academic integrity
8	Zeng, 2024	Classical machine learning	Logistic Regression; SVM; Naive Bayes	Classify text using traditional ML models
9	Malveen, 2024	NLP + ML pipeline	Text pre-processing; feature extraction; classifiers	Detect AI text with NLP and ML methods
10	Prova, 2024	NLP + ML/DL comparison	XGBoost; SVM; BERT	Compare ML and transformer models for detection
11	Fariello et al., 2025	Review	Survey of detection advances	Discuss advances and open challenges
12	Zeng, 2024	Classical machine learning	Machine learning models on labelled text	Study ML models for AI generated text detection

CHAPTER 3
INFORMATION ON SEMINAR

3.1 Overview of AI Generated Text Detection

AI Generated Text Detection is a text classification task in which a software system decides whether a given text was written by a human or generated by an AI model. The system analyses the writing style, vocabulary and statistical patterns of the text and gives a label, usually along with a confidence score. It uses machine learning and natural language processing techniques to examine different features of the text.

These systems are useful in education, research, digital communication and online content verification. They help identify the possible origin of written content and distinguish human-written text from AI-generated text. As AI technology continues to develop, these detection systems are becoming increasingly useful for analysing digital information.

3.2 Large Language Models and AI Generated Text

Large Language Models are trained on huge amounts of text and generate new text by predicting the next word again and again. Because of this, AI-generated text often shows regular patterns, such as smooth sentence structures, repeated phrases, balanced sentence lengths and predictable word choices.

Human writing, in comparison, usually shows more variation and personal style. Detectors try to learn these differences by examining linguistic features, word distribution, sentence complexity and relationships between words. Modern AI models can produce natural and human-like text, which makes accurate detection more challenging.

3.3 Need for AI Generated Text Detection

Detection is needed to protect academic integrity, stop plagiarism, reduce fake news, spam and fake reviews, and maintain trust in online content. As language models continue to improve, manual checking becomes less reliable, so automatic detection systems are required.

AI text detection can support academic evaluation, content moderation and online information verification. It can help identify potentially misleading or automatically generated material on websites and social media platforms.

3.4 Working Principle:

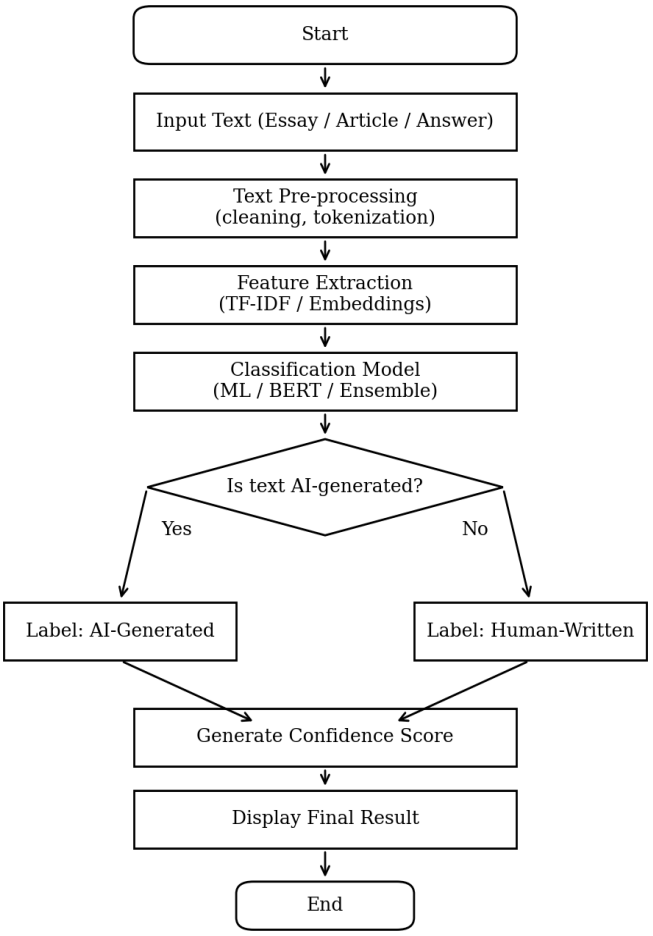


Fig. 3.4: Working of AI-Generated Text Detection [5]

The working of the AI generated text detection system consists of the following steps:

1. Start: The process begins when the user wants to check a piece of text.

2. Input Text: The user enters or uploads the text, such as an essay, article or answer.

3. Text Pre-processing: The text is cleaned by removing extra spaces, symbols and noise, and is split into words or tokens.

4. Feature Extraction: Useful features are taken from the text, such as TF-IDF values, word counts, sentence length, or embeddings produced by a transformer such as BERT.

5. Classification Model: A trained model such as Logistic Regression, SVM, XGBoost, BERT or an ensemble analyses the features and predicts whether the text is human written or AI generated.

6. Decision: Based on the prediction, the text is labelled as AI-Generated or Human-Written.

7. Confidence Score: The system also generates a probability or confidence score for the label.

8. Display Final Result: The label and the score are shown to the user.

9. End: The process ends after the result is delivered.

3.5 System Architecture and Main Components:

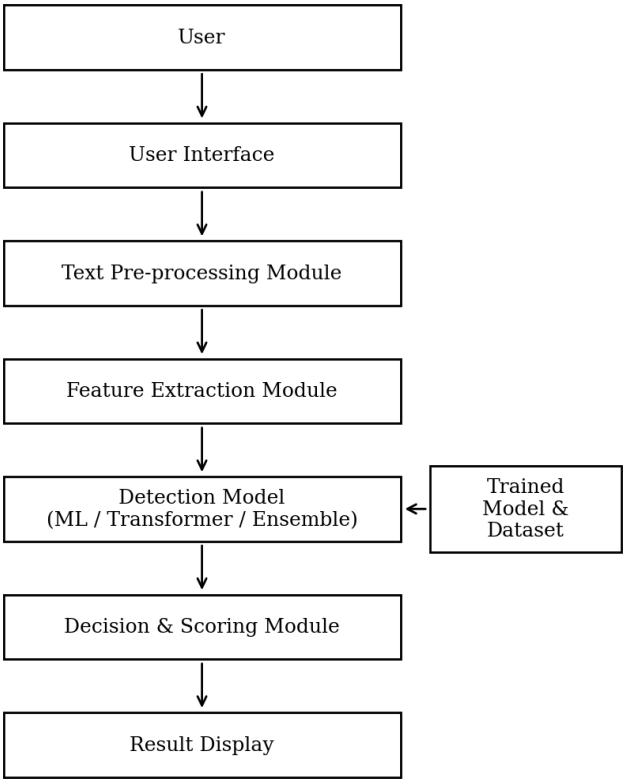


Fig. 3.5: Architecture of AI-Generated Text Detection System [11]

The components are: User Interface, Text Pre-processing Module, Feature Extraction Module, Detection Model, Decision and Scoring Module, and Result Display. The detection model is trained beforehand on a dataset of human written and AI generated text and is then used to classify new text.

3.6 Techniques and Technologies Used:

Python can be used to build the detector. The main techniques seen in the reviewed papers are:

- **Classical machine learning:** Logistic Regression, Support Vector Machine, Naive Bayes and XGBoost, working on features such as TF-IDF.
- **Transformer models:** BERT and similar models that understand the context of words and can be fine-tuned for detection.
- **Ensemble learning:** several models or encoders are combined, sometimes with a gating network, to get better and more robust results.
- **Explainable AI:** methods that show which words or features led to the decision, which is useful in education.

3.7 Performance Evaluation Metrics:

The performance of a detector is measured using accuracy, precision, recall and F1-score. Accuracy shows the overall percentage of correct predictions, precision shows how many texts flagged as AI are really AI generated, recall shows how many AI texts were found, and F1-score balances precision and recall.

3.8 Example of Detection:

A student submits a paragraph. The system pre-processes it, extracts features and passes them to the trained model. The model returns “AI-Generated” with a confidence of, for example, 87%. The teacher can then review the text further instead of depending only on the tool.

3.9 Challenges in Detection:

Detection is difficult because language models are improving quickly, texts may be edited or paraphrased by humans, short texts give little evidence, and a detector trained on one domain may perform poorly on another. Detection results should therefore be treated as an indication and not as final proof.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Helps to protect academic integrity by identifying AI written assignments.
2. Reduces the spread of fake news, spam and fake reviews.
3. Saves the time of teachers and reviewers compared to manual checking.
4. Can work on large amounts of text quickly and automatically.
5. Ensemble and transformer based models improve accuracy and robustness.
6. Explainable AI makes the decisions of the detector easier to understand and trust.

4.2 Disadvantages:

1. Detectors may wrongly mark human written text as AI generated (false positives).
2. Detectors may fail to identify AI text that has been paraphrased or edited.
3. Performance may drop on new domains or on newer language models that the detector was not trained on.
4. Short texts do not give enough information for a reliable decision.
5. Deep learning models need large datasets and more computing power.
6. The result is a probability and cannot be treated as final proof of misconduct.

CHAPTER 5
CONCLUSION

AI Generated Text Detection is an important research area because of the rapid growth of language models and their increasing use in education, research and online content creation. The reviewed studies show that classical machine learning algorithms, transformer models such as BERT, ensemble learning, domain-gating approaches and Explainable Artificial Intelligence (XAI) techniques can help distinguish human-written text from AI-generated text. These techniques analyse writing styles, vocabulary, sentence structures and statistical patterns to identify the possible origin of a given text.

However, several challenges remain in AI-generated text detection, including robustness to paraphrasing, performance across different domains, detection of text generated by newer language models and the risk of false positive predictions. Combining advanced machine learning models with explainable AI techniques can make predictions easier to understand and help users identify the factors influencing the results. Detection systems should be used as supporting tools for human judgement rather than as the sole basis for making important decisions.

Furthermore, developing a practical AI Generated Text Detection System requires appropriate data collection, preprocessing, feature extraction, model training and performance evaluation. Regular testing and model improvements are necessary to maintain reliable performance as AI technology continues to evolve. Overall, this study provides a strong foundation for understanding existing AI text detection techniques, their advantages and limitations, and the challenges involved in their implementation. It also supports the development of an efficient, reliable and practical AI Generated Text Detection System that can assist in identifying potentially AI-generated content and maintaining trust in digital information.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement:

With the growth of AI tools such as ChatGPT, it has become very easy to generate essays, reports and answers. In educational institutions this creates difficulty in checking whether the work submitted by students is original. Manual checking is slow and unreliable because AI generated text looks natural and fluent. Therefore, a system is needed that can automatically analyse a given text and tell whether it is human written or AI generated, along with a confidence score, so that teachers can take informed decisions.

6.2 Scope of the Project:

The purpose of this project is to design an AI Generated Text Detection system that accepts text from the user, pre-processes it, extracts features and uses a trained machine learning model to classify it as human written or AI generated. The system will display the result with a confidence score. The project will focus on English text and on educational content such as essays and answers.

Objectives of the Project:

1. To study existing methods for AI generated text detection.
2. To collect and prepare a dataset of human written and AI generated text.
3. To extract suitable features from the text using NLP techniques.
4. To build a simple interface that shows the prediction and confidence score.

6.3 Stakeholders:

The stakeholders of this project are those who will use the system, develop and maintain it, or be affected by it.

1. **Students:** Users who want to check the originality of their own work.
2. **Teachers and Evaluators:** Users who check assignments for academic integrity.
3. **Developers:** Team members responsible for developing, testing and maintaining the system.
4. **Educational Institutions:** Organizations that want to maintain academic honesty.
5. **System Administrators:** Persons who manage and maintain the system.
6. **Project Guide:** Faculty member who guides the project and evaluates its progress.

6.4 Platform / Equipment / Resources:

1. Platform:

The system can be built on Windows or Linux using the Python programming language. Visual Studio Code can be used for development, and Google Colab can be used if extra computing power is needed for training transformer models.

2. Equipment:

- Computer or Laptop: For developing and running the application.
- Processor: A standard modern processor suitable for Python development.
- RAM: Minimum 8 GB recommended; more may be needed for transformer models.
- Internet Connection: Required for downloading datasets, libraries and models.
- Storage: Sufficient space for source code, datasets and model files.

3. Resources:

Table No. 6.1: Resources[6]

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools, editors and the local testing environment.
Programming Language	Python 3.11	Primary language used to implement the application code, scripts and tests.
Code Editor	Visual Studio Code	Code editing, debugging and extension support for development.
ML Library	scikit-learn	Logistic Regression, SVM, Naive Bayes, TF-IDF and evaluation metrics.
Gradient Boosting	XGBoost	Gradient boosted tree classifier for text features.
Transformer Library	Hugging Face Transformers, PyTorch	Fine-tuning and using BERT for text classification.
Data Handling	pandas, NumPy	Loading, cleaning and processing the dataset.
NLP Library	NLTK / spaCy	Tokenization and text pre-processing.
Dataset	Public human vs AI text datasets	Training and testing the detection models.

CHAPTER 7
SEMINAR PRESENTATION

	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE		
Sem : CO 5 K	**SPI – 315003**	**Date : 17/10/2026**
Topic : AI Generated Text Detection		

Roll No.	Name of Student
67	Mansi Tulshidas Patil
68	Prachi Nilesh Patil
90	Jagruti Baliram Shinde
121	Nikita Laxman Patil

Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6. 1 Description of problem statement.
   - 6. 2 Scope and objectives.
   - 6. 3 Stake holder
   - 6. 4 seminar exsicussion plan
   - 6.5 Platform/ Equipment/ Resources identification.
1. Bibliography
2. References

Abstract

- Large Language Models (LLMs) such as ChatGPT make it very easy to generate fluent, human-like text. This raises concerns in education, journalism and research: plagiarism, misinformation and loss of authenticity.
- AI Generated Text Detection is the task of automatically deciding whether a text was written by a human or produced by a machine.
- This report reviews 12 research papers (2023–2026): classical ML (Logistic Regression, SVM, Naive Bayes, XGBoost), BERT, ensemble and domain-gating approaches, explainable AI, and survey articles.
- It explains the working principle, architecture, techniques, advantages and limitations, and ends with a project initiation plan for a detector that supports academic integrity.

3

1. Introduction

- AI is the field of building machines that perform tasks needing human intelligence; Machine Learning learns patterns from data, and Deep Learning uses multi-layered neural networks.
- Natural Language Processing (NLP) lets computers understand, analyze and generate human language (translation, chatbots, summarization, text classification).
- LLMs such as ChatGPT, Gemini and Claude now write essays, news articles and emails that closely resemble human writing.
- AI-generated text detection combines NLP, ML and DL, using features like word distribution, sentence structure, writing style and perplexity.
- It remains challenging: models keep improving, text can be paraphrased, and human and AI text can look alike.

4

2. Literature Survey

12 research papers (2023–2026) were reviewed:

- **Ensemble / domain-aware:** Pudasaini et al. (2025) – frozen pretrained encoders with ensemble learning; Tripathi et al. (2025) – Domain Gating Ensemble Network.
- **Transformer-based:** Yu (2026) – optimizing BERT; Prova (2024) – compares XGBoost, SVM and BERT.
- **Classical ML and NLP:** Zeng (2024, two papers) – Logistic Regression, SVM, Naive Bayes; Malveen (2024); Alikhanov et al. (2026); Jwaid et al. (2024).
- **Explainable AI:** Najjar et al. (2023) – ML with XAI for academic integrity in education.
- **Review papers:** Lalit et al. (2023); Fariello et al. (2025) – advances and open challenges.

5

Table 2.1: Literature Review (1/2)

No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
1	Pudasaini et al., 2025	Ensemble of frozen encoders	Frozen pretrained encoders; ensemble learning	Detect AI text with good accuracy at low training cost
2	Tripathi et al., 2025	Domain-aware ensemble	Domain Gating Ensemble Network (gating + multiple detectors)	Keep detection accurate across different text domains
3	Yu, 2026	Transformer optimization	BERT; model tuning and optimization	Improve BERT performance for AI text detection
4	Alikhanov et al., 2026	NLP and ML based detection	NLP techniques; machine learning models	Distinguish human and AI generated text
5	Lalit et al., 2023	Review	Survey of existing detection methods	Summarize techniques and challenges
6	Jwaid et al., 2024	Automated detection	Machine learning classification of machine-generated text	Separate human and machine text automatically

6

Table 2.1: Literature Review (2/2)

No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
7	Najjar et al., 2023	ML with Explainable AI	Machine learning classifiers; explainable AI	Detect AI text in education and support academic integrity
8	Zeng, 2024	Classical machine learning	Logistic Regression; SVM; Naive Bayes	Classify text using traditional ML models
9	Malveen, 2024	NLP + ML pipeline	Text pre-processing; feature extraction; classifiers	Detect AI text with NLP and ML methods
10	Prova, 2024	NLP + ML/DL comparison	XGBoost; SVM; BERT	Compare ML and transformer models for detection
11	Fariello et al., 2025	Review	Survey of detection advances	Discuss advances and open challenges
12	Zeng, 2024	Classical machine learning	Machine learning models on labelled text	Study ML models for AI generated text detection

7

3. Information to seminar topic

- **Overview:** a text classification task that labels text as human-written or AI-generated, with a confidence score.
- **AI text patterns:** smooth sentences, repeated phrases, balanced lengths, predictable word choices; human writing varies more.
- **Need:** academic integrity, plagiarism, fake news, spam and fake reviews, trust in online content.
- **Working principle:** Input text → Pre-processing → Feature extraction (TF-IDF / embeddings) → Classification model (ML / BERT / Ensemble) → Label + confidence score → Display result.
- **Architecture:** User Interface, Pre-processing Module, Feature Extraction Module, Detection Model (trained on a dataset), Decision & Scoring Module, Result Display.
- **Techniques:** Classical ML (LR, SVM, Naive Bayes, XGBoost), Transformers (BERT), Ensemble learning, Explainable AI.
- **Evaluation metrics:** accuracy, precision, recall and F1-score.
- **Challenges:** paraphrased text, short texts, new domains and newer LLMs; a result is an indication, not final proof.

8

4. Advantages / Disadvantages

Advantages

- Protects academic integrity by identifying AI-written assignments
- Reduces fake news, spam and fake reviews
- Saves teachers' and reviewers' time compared to manual checking
- Works quickly and automatically on large amounts of text
- Ensemble and transformer models improve accuracy and robustness
- Explainable AI makes decisions easier to understand and trust

Disadvantages

- False positives: human text may be marked as AI-generated
- May fail on paraphrased or edited AI text
- Performance drops on new domains or newer language models
- Short texts do not give enough information
- Deep learning needs large datasets and more computing power
- Result is a probability, not final proof of misconduct

9

5. Conclusion

- AI Generated Text Detection is an important research area because of the rapid growth of language models in education, research and online content.
- Classical ML, BERT, ensemble learning, domain-gating and Explainable AI can all help separate human-written text from AI-generated text.
- Challenges remain: robustness to paraphrasing, different domains, newer language models and false positives.
- Detectors should support human judgement, not be the sole basis for important decisions.
- A practical system needs data collection, preprocessing, feature extraction, model training, evaluation and regular improvement.

10

6. Project Initiation

- **Project:** AI Generated Text Detection System
- Aim: a system that accepts text from the user, pre-processes it, extracts features and uses a trained ML model to classify it as human-written or AI-generated, with a confidence score.

This section covers:

- 6.1 Description of problem statement
- 6.2 Scope and objectives
- 6.3 Stakeholders
- 6.4 Seminar discussion plan
- 6.5 Platform / Equipment / Resources

11

6. 1 Description of problem statement

- Tools such as ChatGPT make it very easy to generate essays, reports and answers.
- In educational institutions it is difficult to check whether submitted work is original; manual checking is slow and unreliable because AI text looks natural and fluent.
- Therefore, a system is needed that automatically analyses a text and tells whether it is human-written or AI-generated, with a confidence score, so teachers can take informed decisions.

12

6. 2 Scope and objectives.

Scope

- Design a system that accepts text from the user, pre-processes it, extracts features and classifies it as human-written or AI-generated using a trained ML model.
- Displays the result with a confidence score.
- Focus on English text and educational content such as essays and answers.

Objectives

- To study existing methods for AI-generated text detection
- To collect and prepare a dataset of human-written and AI-generated text
- To extract suitable features using NLP techniques
- To build a simple interface showing the prediction and confidence score

13

6. 3 Stake holder

- **Students:** check the originality of their own work.
- **Teachers and Evaluators:** check assignments for academic integrity.
- **Developers:** develop, test and maintain the system.
- **Educational Institutions:** want to maintain academic honesty.
- **System Administrators:** manage and maintain the system.
- **Project Guide:** faculty member who guides the project and evaluates its progress.

14

6. 4 seminar exsicussion plan

- Introduction – AI, NLP and Large Language Models
- Need for AI generated text detection
- Literature review – 12 research papers (2023–2026) and comparison table
- Working principle and system architecture
- Techniques, technologies and performance evaluation metrics
- Advantages, disadvantages and challenges
- Conclusion and project initiation (problem statement, scope, stakeholders, resources)
- Question and answer session

15

6.5 Platform/ Equipment/ Resources

Platform

- Windows or Linux, Python programming language
- Visual Studio Code for development
- Google Colab if extra computing power is needed for transformer models

Equipment

- Computer or laptop with a standard modern processor
- RAM: minimum 8 GB recommended (more for transformer models)
- Internet connection for datasets, libraries and models
- Sufficient storage for source code, datasets and model files

16

Table 6.1: Resources

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools, editors and the local testing environment.
Programming Language	Python 3.11	Primary language used to implement the application code, scripts and tests.
Code Editor	Visual Studio Code	Code editing, debugging and extension support for development.
ML Library	scikit-learn	Logistic Regression, SVM, Naive Bayes, TF-IDF and evaluation metrics.
Gradient Boosting	XGBoost	Gradient boosted tree classifier for text features.
Transformer Library	Hugging Face Transformers, PyTorch	Fine-tuning and using BERT for text classification.
Data Handling	pandas, NumPy	Loading, cleaning and processing the dataset.
NLP Library	NLTK / spaCy	Tokenization and text pre-processing.
Dataset	Public human vs AI text datasets	Training and testing the detection models.

17

Bibliography

- **Books**
  - Artificial Intelligence: A Modern Approach — Stuart Russell and Peter Norvig
  - Speech and Language Processing — Daniel Jurafsky and James H. Martin
  - Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow — Aurelian Geron.
  - Natural Language Processing with Transformers — Lewis Tunstall, Leandro von Werra and Thomas Wolf
- **Websites**
  - [Google Scholar](https://scholar.google.com/?utm_source=chatgpt.com): – Research papers related to AI-Generated Text Detection.
  - [ResearchGate](https://www.researchgate.net/?utm_source=chatgpt.com): – Research articles on Human vs. Machine-Generated Text.
  - [IEEE Xplore](https://ieeexplore.ieee.org/?utm_source=chatgpt.com): – Technical research papers on AI and Machine Learning.
  - [ScienceDirect](https://www.sciencedirect.com/?utm_source=chatgpt.com) :– Articles related to AI-Generated Text Detection.

18

References

[1] S. Pudasaini, L. Miralles-Pechuán, D. Lillis and M. Llorens Salvador, "Enhancing AI Text Detection with Frozen Pretrained Encoders and Ensemble Learning," 2025.

[2] A. Tripathi, E. Jin, P. Zhang et al., "Domain Gating Ensemble Networks for AI-Generated Text Detection," 2025.

[3] J. Yu, "Research on Performance Optimization of BERT Model in AI-Generated Text Detection," 2026.

[4] A. Alikhanov, D. Akhmetzhan, A. Amangeldi et al., "AI Generated Text Detection," 2026.

[5] R. Lalit, P. Bhutani, N. Verma and A. Jain, "AI-Generated Text Detection: A Review," 2023.

[6] U. Jwaid, R. Roy, P. Pal et al., "Human vs Machine: An Automated Machine-Generated Text Detection Approach," 2024.

[7] A. A. Najjar, H. I. Ashqar, O. A. Darwish and E. Hammad, "Detecting AI-Generated Text in Educational Content: Leveraging Machine Learning and Explainable AI for Academic Integrity," 2023.

[8] M. Zeng, "AI-Generated Text Detection Based on Machine Learning Models," 2024.

[9] Z. Malveen, "Detecting AI-Generated Text Through NLP and Machine Learning Methodologies," 2024.

[10] N. N. I. Prova, "Detecting AI Generated Text Based on NLP and Machine Learning Approaches," 2024.

[11] S. Fariello, G. Fenza, F. Forte, M. Gallo and M. Marotta, "Distinguishing Human From Machine: A Review of Advances and Challenges in AI-Generated Text Detection," 2025.

[12] M. Zeng, "Research on AI-Generated Text Detection Based on Machine Learning Models," 2024.

Thank you

BIBLIOGRAPHY

- **Books**

1. Artificial Intelligence: A Modern Approach — Stuart Russell and Peter Norvig.
2. Speech and Language Processing — Daniel Jurafsky and James H. Martin.
3. Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow — Aurelian Geron.
4. Natural Language Processing with Transformers — Lewis Tunstall, Leandro von Werra and Thomas Wolf.

- **Websites**

1. [Google Scholar] (https://scholar.google.com/?utm_source=chatgpt.com) – Research papers related to AI-Generated Text Detection.
2. [ResearchGate] (https://www.researchgate.net/?utm_source=chatgpt.com) – Research articles on Human vs. Machine-Generated Text.
3. [IEEE Xplore] (https://ieeexplore.ieee.org/?utm_source=chatgpt.com) – Technical research papers on AI and Machine Learning.
4. [ScienceDirect] (https://www.sciencedirect.com/?utm_source=chatgpt.com) – Articles related to AI-Generated Text Detection.

REFERENCES

1. S. Pudasaini, L. Miralles-Pechuán, D. Lillis and M. Llorens Salvador, “Enhancing AI Text Detection with Frozen Pretrained Encoders and Ensemble Learning,” 2025.
2. A. Tripathi, E. Jin, P. Zhang, L. Dugan, C. Gao, D. Zhang, J. Zhao, M. Huan and C. Callison-Burch, “Domain Gating Ensemble Networks for AI-Generated Text Detection,” 2025.
3. J. Yu, “Research on Performance Optimization of BERT Model in AI-Generated Text Detection,” 2026.
4. A. Alikhanov, D. Akhmetzhan, A. Amangeldi, O. Polat, D. Demeubay, G. Zharas and N. Moldakhmetov, “AI Generated Text Detection,” 2026.
5. R. Lalit, P. Bhutani, N. Verma and A. Jain, “AI-Generated Text Detection: A Review,” 2023.
6. U. Jwaid, R. Roy, P. Pal, S. Debnath, D. Das and S. Bandyopadhyay, “Human vs Machine: An Automated Machine-Generated Text Detection Approach,” 2024.
7. A. A. Najjar, H. I. Ashqar, O. A. Darwish and E. Hammad, “Detecting AI-Generated Text in Educational Content: Leveraging Machine Learning and Explainable AI for Academic Integrity,” 2023.
8. M. Zeng, “AI-Generated Text Detection Based on Machine Learning Models,” 2024.
9. Z. Malveen, “Detecting AI-Generated Text Through NLP and Machine Learning Methodologies,” 2024.
10. N. N. I. Prova, “Detecting AI Generated Text Based on NLP and Machine Learning Approaches,” 2024.
11. S. Fariello, G. Fenza, F. Forte, M. Gallo and M. Marotta, “Distinguishing Human From Machine: A Review of Advances and Challenges in AI-Generated Text Detection,” 2025.
12. M. Zeng, “Research on AI-Generated Text Detection Based on Machine Learning Models,” 2024.

SEMINAR and PROJECT INITIATION Assessment / Evaluation

Formative Assessment

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960248	388554	Mansi Tulshidas Patil
2	24610960249	388555	Prachi Nilesh Patil
3	24610960276	388577	Jagruti Baliram Shinde
4	25610960010	388608	Nikita Laxman Patil

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388554	388555	388577	388608
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name: Mr. Nishad D. Patel

(Course Expert/s)